

EDUCATION

University of California, Berkeley

Berkeley, CA

*Bachelors in Computer Science, Mathematics, Statistics (Triple Major) GPA: 3.9;**Jun 2020 – May 2024*

Organizations: CS Honor Society Officer, CSUA (CS Club Executive Officer), CSM (CS 61A/B/70/88 Tutor)

Relevant Coursework: Data Structures, Algorithms, Probability & Stochastic Processes[†], Signal Processing, Networking, Linear Algebra, Parallel Systems[†], Full Stack Web[†], Comp. Arch, Multivariable Calc., Compilers, Robotics[†], Comp. Graphics[†], Comp. Security, UI/UX, Data Science & Machine Learning, Neural Nets[†], Deep Reinforcement Learning[†], Natural Language Processing[†], Statistical Forecasting[†], Comp. Vision*, Causal Inference*, Real Analysis*, Data Mining*, Numerical Analysis*, Financial Engineering*, Theoretical Statistics*[†] [†]= Graduate Coursework * = In Progress

EXPERIENCE

• **Schonfeld Quantitative Hedge Fund**

New York, NY

*Current Alpha Quantitative Researcher Intern**Apr 2023 - Aug 2023*

- Exploratory Data Analysis using Databricks' PySpark over 100 GB of alternative data from AWS S3 and EC2, with storage optimized via Parquet files, for a statistically backtested signal. Under **Systematic Trading** team (QIS).
- Wrote an Online Algorithm to improve latency by 50% via Statistical Estimation for a **HFT Stat Arb** strategy. Measured effects of out-of-sample distributional shift via divergence measures. Wrote scheduler to ingest live data.

• **Arrowstreet Capital LP**

Boston, MA

*Quantitative Developer Intern @ Arrowstreet's Research Core Team**Jun 2022 - Aug 2022*

- Wrote distributed ML algorithms to discover hidden signals for equity alpha research using ~100,000 vendor filings. Robustly improved the firm-wide single systematic process with Bayesian regressions & other new equities toolings.

• **University of California, Berkeley**

Berkeley, CA

*Teaching: Head TA, uGSI, Tutor, Reader; Researcher, Research Assistant**Jan 2021 - Present*

- Head Teaching Assistant - CS C100: Data Science (Pandas) and Machine Learning:** Managed admin logistics for ~300 students related to the EdStem communication platform, assignment extensions (automated using Gradescope API), Incomplete Student assignment linkages and DSP communications, held OH and taught sections
- Lead Teaching Assistant - EECS 122: Communication Networking/Internet:** Managed all administrative course logistics, created all existing course content, held office hours and **taught section** on-campus
- Reader, Academic Student Employee:** Graded (and wrote) exams on Data Structures (Lists/HashMaps/Heaps/Trees/etc) & Discrete Math (Logic, Crypto, Markov Chains, etc) & held Office Hours for 800+ students
- Research - Artificial Intelligence:** Research in Deep Learning & NLP: Trained an LSTM to predict student grades from attendance and conducted statistical experiments to determine its effectiveness (Published a paper to be presented at ACM's 2023 conference). Fine-tuned a LLM on past TA messages on the EdStem Q&A Platform.

PROJECTS

- GPU-accelerated Factorization Machines:** Boosted recommender systems training speed by **21.4%** using a novel randomized algorithm that capitalized on data sparsity to parallelize Stochastic Gradient Descent. Achieved via custom CUDA structures for efficient low-rank polynomial kernel approximation. Profiled bottlenecks with VTune's Roofline.
- Berkeley AI Research (BAIR) Lab:** Collaborated with Simeon Adebola (PhD candidate) and Professor Goldberg from AUTOLab to develop a robust automated cable insertion system for dynamic environments. Applied CV and flood-fill algorithms for cable and channel identification, using perception, segmentation, tracing, & endpoint detection.
- RiskWatch:** Developed React.js Application for Chubb; winner of PennApps 18, using AWS Rekognition to classify safety hazards from a Raspberry Pi camera. Top 10 project addressing risk/disaster by UPenn's Wharton Risk Center

AWARDS

- Donald M. Rizzo Memorial Scholarship Recipient:** Awarded from international NPO: Lions Club International
- Winner of the Congressional App Challenge:** Invited to Capitol Hill to demo mobile app to House of Representative members at #HouseOfCode, the most prestigious student STEM expo, at the U.S. Capitol Building
- Capture The Flag Cybersecurity Competitions:** Ranked between top 5 to top 50 at all of {PicoCTF, HSCTF, BRACTF, CSAW Red}: Security Competitions with challenges from programming questions to hacking into servers

SKILLS

- Languages:** Python, Java, C++, C, CUDA, SQL, JavaScript, React, Rust, OCaml, Stata, R, Go, x86, Lisp, & C#
- Technologies:** Pandas, statsmodels, sklearn, Matplotlib/Seaborn, git, Scipy, cvxpy, PyTorch, Excel, RegEx, L^AT_EX
- Hobbies:** Hiking, Chess, Reading, Matt Levine, LLMs, Trading/Predicting Events, Escape Room, Traveling, Pokemon